



Zeyu Wang

Nationality: Chinese **Date of birth:** 12/08/1996 **Gender:** Male

Phone number: (+86) 18640180908

Email address: zeyu.wang@mik.uni-pannon.hu

Website: <https://zeyu-w.github.io>

Work: Egyetem u. 10 Faculty of Information Technology, 8200 Veszprém (Hungary)

EDUCATION AND TRAINING

Ph.D. in Computer Science

University of Pannonia [01/09/2021 – 13/08/2025]

City: Veszprém | Country: Hungary | Website: <https://uni-pannon.hu/>

Master in Biomedical Engineering

Shenyang University of Technology [01/09/2018 – 29/06/2021]

City: Shenyang | Country: China | Website: <https://www.sut.edu.cn/index.jsp>

Bachelor in Biomedical Engineering

Shenyang University of Technology [23/08/2014 – 17/06/2018]

City: Shenyang | Country: China | Website: <https://www.sut.edu.cn/index.jsp>

RESEARCH INTERESTS

Substantive

Computational neuroscience, Biomedical signal processing, Brain-computer interface

Methodological:

Parallel computing, High-performance computing, Machine learning

PUBLICATIONS

[2023]

[GPU Implementation of the Improved CEEMDAN Algorithm for Fast and Efficient EEG Time-Frequency Analysis](#)

Z. Wang and Z. Juhasz, "GPU Implementation of the Improved CEEMDAN Algorithm for Fast and Efficient EEG Time-Frequency Analysis", MDPI Sensors (Basel), vol. 23, no. 20, p. 8654, Oct. 2023

[2023]

[Efficient GPU implementation of the multivariate empirical mode decomposition algorithm](#)

Z. Wang and Z. Juhasz, "Efficient GPU implementation of the multivariate empirical mode decomposition algorithm", Journal of Computational Science, vol. 74, p. 102180, Dec. 2023

[2023]

[Classifying epileptic phase-amplitude coupling in SEEG using complex-valued convolutional neural network](#)

C. Li, S. Liu, Z. Wang, G. Yuan, "Classifying epileptic phase-amplitude coupling in SEEG using complex-valued convolutional neural network", Frontiers in Physiology, vol. 13, p. 1085530, Jan. 2023

[2023]

[Massively Parallel EEG Algorithms for Pre-exascale Architectures](#)

Z. Wang and Z. Juhasz, "Massively Parallel EEG Algorithms for Pre-exascale Architectures", European Conference on Parallel Processing (Euro-Par 2023), Limassol, Cyprus, Aug 28 - Sept 1, 2023

[2022]

On the Benefits of Empirical Mode Decomposition in Spatio-temporal EEG Analysis

Z. Wang, Z. Nagy, Z. Juhasz, "On the Benefits of Empirical Mode Decomposition in Spatio-temporal EEG Analysis", 45th Jubilee International Convention on Information, Communication and Electronic Technology, Opatija, Croatia, May 23 - 27, 2022

[2021]

Phase-Amplitude Coupling and Epileptogenic Zone Localization of Frontal Epilepsy Based on Intracranial EEG

H. Ma, Z. Wang, C. Li, J. Chen, Y. Wang, "Phase-Amplitude Coupling and Epileptogenic Zone Localization of Frontal Epilepsy Based on Intracranial EEG", Frontiers in Neurology, vol. 12, p. 718683, Sept 2021

[2025]

Enhancing single-lead electrocardiogram arrhythmia detection with empirical mode decomposition

M.F. Issa, A. Yousry, G. Tuboly, Z. Wang, Z. Juhasz, M.M. Selim, A.H. Abuei-Atta, "Enhancing single-lead electrocardiogram arrhythmia detection with empirical mode decomposition", Neural Computing and Applications, vol. 37, pp. 1-23, Mar. 2025

HONOURS AND AWARDS

[01/09/2021] China Scholarship Council & Hungary Tempus Public Foundation
Stipendium Hungaricum Scholarship Program

[05/07/2023] Euro-Par Foundation
Euro-Par 2023 PhD Studentship

[01/09/2023] Hungary National Research, Development and Innovation Office (NRDI)
New National Excellence Programme (ÚNKP)

[23/11/2024] 2024 International Conference on Science and Engineering of Electronics (ICSEE'2024)
Best oral presentation award

CONFERENCES AND SEMINARS

[22/11/2024 – 24/11/2024] Wuhan, China
2024 International Conference on Science and Engineering of Electronics (ICSEE 2024)
Oral presentation

[25/06/2024 – 29/06/2024] Vienna, Austria
2024 Federation of European Neuroscience Societies Forum (FENS 2024)
Poster presentation

[25/01/2024 – 26/01/2024] Pécs, Hungary
2024 International Neuroscience Conference (INC 2024)
Poster presentation

[28/08/2023 – 01/09/2023] Limassol, Cyprus
29th International European Conference on Parallel and Distributed Computing (Euro-Par 2023)
Oral presentation

[23/05/2022 – 27/05/2022] Opatija, Croatia

45th Jubilee International Convention on Information, Communication and Electronic Technology (IEEE-MIPRO 2022)

Oral presentation

[20/06/2022 – 31/05/2024] Budapest, Hungary

12th, 13th, and 14th GPU Day International Conference

Oral presentation

[03/12/2021 – 07/12/2024] Veszprém & Szeged, Hungary

XXXIV, XXXV, XXXVI, XXXVII. Neumann Colloquium

Oral presentation

[20/07/2020 – 24/07/2020] Montreal, Canada (online)

42nd Annual International Conferences of the IEEE Engineering in Medicine (IEEE-EMBC 2020)

Oral presentation

[24/08/2018 – 27/08/2018] Shenyang, China

2018 IEEE International Conference on Intelligence and Safety for Robotics (IEEE-ISR 2018)

Oral presentation

NETWORKS AND MEMBERSHIPS

[07/02/2024 – Current] Brussels

Member of Federation of European Neuroscience Societies

Link: <https://www.fens.org/>

[31/12/2023 – Current] Budapest

Member of Hungarian Neuroscience Society (MITT)

Link: <https://www.mitt.hu/en>

[03/12/2024 – Current] Budapest

Member of John von Neumann Computer Society (NJSZT)

TEACHING EXPERIENCE

Faculty of Information Technology, University of Pannonia

Teaching Assistant to Dr. Zoltan Juhasz

- GPU Programming 2023/24/1
- GPU Programming 2024/25/1